

Kun-Yu Lee

Evanston, IL · kun-yulee2027@u.northwestern.edu · LinkedIn · GitHub

Research Interests

Evaluating how LLM-based recommenders make decisions under uncertainty: when they should ask, what they surface, and whether their behavior can be audited.

Education

Northwestern University

M.S. in Machine Learning and Data Science

Sep 2025 – Expected Dec 2026

Evanston, IL

- GPA: 3.94/4.00

University of Nebraska–Lincoln

B.S. in Computer Science and B.A. in Data Science; Minor in Mathematics

Aug 2021 – Dec 2024

Lincoln, NE

- GPA: 3.9/4.0; completed two degrees and a minor in 3.5 years.

Publications and Manuscripts

- [1] **Kun-Yu Lee** and Edward C. Malthouse. “When Is a Clarifying Question Worth Asking? Request-Dependent Clarification in LLM-Based Conversational Recommendation.” *To be submitted to the European Conference on Information Retrieval (ECIR 2027), short paper track.*
- [2] Edward C. Malthouse, **Kun-Yu Lee**, Jing Yang, Sanchary Pal, and Xueyan Feng. “The Silent Gatekeeper: A Framework for Measuring, Explaining, and Diagnosing Brand Visibility in LLM Recommendations.” *Under review at the Journal of Advertising Research (JAR), 2026.*
- [3] Edward C. Malthouse, **Kun-Yu Lee**, Jing Yang, Sanchary Pal, and Xueyan Feng. “Evaluating Brand Retrieval and Ranking in Open-Ended LLM Recommendations.” *To be submitted to the European Conference on Information Retrieval (ECIR 2027).* Extended version: “Evaluating Brand Retrieval and Ranking in Large Language Model Recommendations,” *arXiv preprint*, 2026.
- [4] Jing Yang, **Kun-Yu Lee**, et al. “The Machine Brand Image: Evaluating LLM Recommendations Across Prompts and Models.” *To be submitted to the American Academy of Advertising (AAA) 2027 Conference.*

Research Experience

Graduate Research Assistant, Northwestern University

Advisors: Edward C. Malthouse and Jing Yang

Jan 2026 – Present

Evanston, IL

When Should an LLM Recommender Ask? (first author)

Paper [1]

- Designed an exact-oracle benchmark that determines, for each request, whether a clarifying question could change the optimal recommendation.
- Across three frontier LLMs, 77–93% of clarifying questions asked in decision-determined cases could not change the optimal recommendation, even though the models recognized this when asked directly; stating the decision criterion in the prompt largely eliminated these questions.
- Ongoing: extending the oracle to request-dependent clarification on a real-world hotel catalog.

Auditing Brand Recommendations in LLMs (second author)

Papers [2]–[4]

- Contributed to research framing and experimental design for a framework that audits open-ended LLM brand recommendations as a stochastic retrieval-and-ranking process: six commercial LLMs, five categories, repeated stateless sampling, and BRP@ k / MRR@ k for recommendation prevalence and prominence.
- Ran the evaluation experiments, organized and analyzed results, and co-wrote parts of the manuscripts.
- Found that category-only queries omit many established brands and that prominence tracks search interest and online brand conversation more than conventional brand popularity; needs-based queries change which brands are retrieved, and NDCG need-matching varies widely across positioning dimensions.
- Characterized the “machine brand image” from 17,821 LLM justifications coded into eight association types: price tier is its strongest marketplace correlate, while advertising spending shows no association.

Earlier Research, University of Nebraska–Lincoln

Nov 2023 – May 2025

Undergraduate Research Assistant

Lincoln, NE

- *Viral genomics* (Advisor: Qiuming Yao): built an interactive tool to study how mutation patterns vary across virus lineages, regions, and time.
- *EV adoption* (Advisor: Jason Fraser Hawkins): modeled how infrastructure, geography, and time relate to consumer EV adoption using interpretable regression and tree-based models.

Selected Projects

Interpretable LLM Agent for Structured Scouting and Ranking

Hackathon Third Place, Nov 2025

- Built an agentic workflow that applies explicit normalization, constraint, and conflict-resolution rules before ranking candidates; generated rationales for human review and tested ranking stability.

Predicting Real Estate Prices with Limited Local Data

Course project, Dec 2024

- Studied cross-county price prediction under distribution shift; in this setting, models trained on local-only data had 61–69% lower MSE than models trained on larger pooled data.

Honors and Awards

Third Place, Northwestern MLDS AI Hackathon (see Selected Projects)

Nov 2025

Dean’s Commendation List (Fall 2024); Dean’s List (3 semesters), University of Nebraska–Lincoln

2021–2024

Additional Experience

Financial Trading Assistant, CTBC Financial Holding

Jun 2023 – Aug 2023

Market-data feature engineering and visualization

Taipei, Taiwan

- Engineered real-time quantitative features (price, volatility) from high-frequency Bloomberg data and built dashboards used by traders and analysts.

Learning Assistant, Department of Mathematics, University of Nebraska–Lincoln

Aug 2023 – Dec 2023

Undergraduate mathematics

Lincoln, NE

- Led weekly recitations and office hours and helped students build mathematical reasoning for modeling and data analysis.